

# FTS Midterm Exam IV (Weeks 2–5)

Oct 8, 2026 · @Keehong Rhee

## Answer Key (Instructor Copy — remove before printing)

Weeks 2–5 only. Answers are balanced at 10 each for A, B, C and D. Q20–24 form the AR ↔ MA block. Detailed solutions for review follow the table.

| Q  | Ans | Week | Reason                                                                                                                                   |
|----|-----|------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | B   | W2   | Prices wander; returns keep a stable average and spread.                                                                                 |
| 2  | D   | W2   | Weak stationarity: constant mean, variance and lag structure.                                                                            |
| 3  | A   | W2   | A growing spread breaks the constant-variance condition.                                                                                 |
| 4  | C   | W2   | $ \phi  < 1$ mean-reverts; $\phi = 1$ is a unit root.                                                                                    |
| 5  | C   | W2   | Random-walk variance grows with time ( $t\sigma^2$ ).                                                                                    |
| 6  | A   | W2   | The first difference of a random walk is white noise.                                                                                    |
| 7  | D   | W2   | Slow ACF decay is the unit-root signature.                                                                                               |
| 8  | B   | W2   | Detrend a fixed trend; difference a unit root.                                                                                           |
| 9  | A   | W2   | A large negative lag-1 correlation signals over-differencing.                                                                            |
| 10 | B   | W2   | Stationary gap between $I(1)$ series = cointegration.                                                                                    |
| 11 | D   | W2   | $z = (16 - 10)/2 = +3$ : short the rich leg, long $\beta$ units of the cheap leg.                                                        |
| 12 | D   | W2   | Cointegration can break; with the leash gone there is no reason to expect reversion, so exit (stop-loss).                                |
| 13 | A   | W3   | AR echo decays as $\phi$ to the power $h$ .                                                                                              |
| 14 | C   | W3   | MA(1) memory ends after one lag.                                                                                                         |
| 15 | B   | W3   | ACF = total link; PACF = direct link.                                                                                                    |
| 16 | D   | W3   | ACF cuts off at $q = 2 \rightarrow$ MA(2).                                                                                               |
| 17 | B   | W3   | PACF cuts off at $p = 1 \rightarrow$ AR(1).                                                                                              |
| 18 | A   | W3   | Both fade $\rightarrow$ ARMA; start small.                                                                                               |
| 19 | C   | W3   | AR(1) is stationary only if $ \phi  < 1$ .                                                                                               |
| 20 | C   | W3   | AR $\rightarrow$ MA: stationary AR(1) = MA( $\infty$ ) with weights $\phi$ to the power $j$ (Wold).                                      |
| 21 | D   | W3   | Why AR $\rightarrow$ MA: it shows how a shock fades over time; shrinking weights mean stationary.                                        |
| 22 | B   | W3   | MA $\rightarrow$ AR: invertible MA(1) = AR( $\infty$ ) with weights $-(-\theta)^j$ : 0.5, -0.25, 0.125, ...                              |
| 23 | A   | W3   | Why MA $\rightarrow$ AR: shocks are unobserved; the AR( $\infty$ ) form recovers them from data for forecasting (needs $ \theta  < 1$ ). |

| Q  | Ans | Week | Reason                                                                          |
|----|-----|------|---------------------------------------------------------------------------------|
| 24 | C   | W3   | AR and MA describe the same process two ways; pick the parsimonious form.       |
| 25 | B   | W3   | Stationary forecasts revert to the mean; the band widens.                       |
| 26 | A   | W3   | Leftover residual structure → revise the model.                                 |
| 27 | D   | W4   | ARIMA = difference, fit ARMA, integrate back.                                   |
| 28 | B   | W4   | Random walk = ARIMA(0,1,0).                                                     |
| 29 | A   | W4   | Stationary forecasts revert; unit-root forecasts continue with a widening band. |
| 30 | C   | W4   | Choose the smallest d that reaches stationarity.                                |
| 31 | D   | W4   | ACF spikes at multiples of 12 = yearly season.                                  |
| 32 | A   | W4   | Seasonal difference $(1 - L^{12})$ removes the yearly pattern.                  |
| 33 | C   | W5   | Direction has no memory; size does (volatility clustering).                     |
| 34 | D   | W5   | Conditional variance changes with recent shocks.                                |
| 35 | B   | W5   | ARCH squares the shock: only the size matters.                                  |
| 36 | A   | W5   | $\beta$ = memory of past variance (persistence).                                |
| 37 | D   | W5   | Persistence closer to 1 → longer half-life.                                     |
| 38 | B   | W5   | Volatility mean-reverts to its long-run level.                                  |
| 39 | C   | W5   | VaR is a threshold breached on about 1% of days.                                |
| 40 | C   | W5   | Fixed VaR ignores changing risk; GARCH VaR adapts.                              |

## Detailed Solutions (Review Guide)

Each solution gives three things: why the correct answer is right, why each wrong option fails, and the one idea to take away. Work through the question first, then read the solution.

### Section I • Does the Series Stay Put?

#### Q1 • Answer B

**Why B.** A stationary series has a fixed average and a fixed size of swings. The price drifts upward and never settles, so its average changes over time. The daily returns stay centred on zero with similar swings throughout, so they meet the conditions.

**Why not the others.** A: a drifting price has no fixed level to return to. C: only the returns pass. D: the returns do pass.

**Takeaway.** In finance, prices usually wander, while returns (the changes) are usually close to stationary. That is why we model returns.

## Q2 · Answer D

**Why D.** Weak stationarity asks for three things to stay the same over time: the mean, the variance, and the autocovariance at each lag  $k$ .

$$E[X_t] = \mu, \quad \text{Var}(X_t) = \sigma^2, \quad \text{Cov}(X_t, X_{t-k}) = \gamma_k \text{ (depends only on } k\text{)}$$

**Why not the others.** A: a stationary series moves a lot; it just moves around a stable level. B: a constant rise is a trend, which changes the mean. C: stationarity says nothing about the shape of the distribution.

**Takeaway.** Stationary means "the rules of the game do not change", not "nothing moves".

## Q3 · Answer A

**Why A.** The variance must be constant. Swings that grow every year mean the variance grows, so the series fails even though its average is steady.

**Why not the others.** B: a constant mean is necessary but not enough. C: a bell shape is irrelevant. D: stationarity is a property of the series, not of a forecast.

**Takeaway.** Check all three conditions — mean, spread and lag links. Failing any one breaks stationarity.

## Q4 · Answer C

**Why C.** In the AR(1),  $\phi$  is the share of yesterday's deviation that survives into today. With  $\phi = 0.5$ , half of any deviation disappears each day, so the series is pulled back to its mean, like a dog on a leash. With  $\phi = 1$ , nothing disappears:  $X_t = X_{t-1} + \varepsilon_t$  is a random walk and every shock stays forever.

$$|\phi| < 1 : \text{Var}(X_t) = \frac{\sigma^2}{1 - \phi^2} \text{ (finite)} \quad \phi = 1 : X_t = X_{t-1} + \varepsilon_t \text{ (no fixed variance)}$$

**Why not the others.** A: only  $\phi = 0.5$  pulls back. B: only  $\phi = 1$  is a random walk. D:  $\phi = 1$  means no pull at all — it never returns on purpose.

**Takeaway.** The closer  $\phi$  is to 1, the looser the leash. At  $\phi = 1$  the leash is gone.

## Q5 · Answer C

**Why C.** A random walk that starts at zero is the sum of all past shocks. Every day adds one more shock and none fade, so the variance adds up.

$$X_t = \varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_t \quad \Rightarrow \quad \text{Var}(X_t) = t \sigma^2$$

After 100 days the variance is 10 times that after 10 days, so the spread (standard deviation) is about  $\sqrt{10} \approx 3.2$  times wider.

**Why not the others.** A: without drift the average stays at zero; only the spread grows. B: fat tails are about the shape, not the growing spread. D: this is exactly what theory predicts.

**Takeaway.** For a random walk, uncertainty grows with the horizon (variance  $\propto t$ ). That is why it is not stationary.

## Q6 · Answer A

**Why A.** Subtract yesterday from today:

$$X_t = X_{t-1} + \varepsilon_t \quad \Rightarrow \quad \Delta X_t = X_t - X_{t-1} = \varepsilon_t$$

What is left is the shock itself — white noise, which is stationary.

**Why not the others.** B, C: differencing removes a trend; it does not create one or make the series explode. D: the difference is no longer a random walk.

**Takeaway.** Differencing is the cure for a unit root: the difference of a random walk is white noise.

## Q7 · Answer D

**Why D.** A unit-root series remembers almost everything, so its ACF starts near 1 and decays very slowly. Returns have almost no memory, so their ACF drops to about zero right after lag 0. The price should be differenced (which gives returns); the returns already look stationary.

**Why not the others.** A: it is the other way round. B: the returns show no unit-root sign. C: a slow ACF reflects a wandering level, not a profit opportunity.

**Takeaway.** A slowly decaying ACF is the visual signature of a unit root.

## Q8 · Answer B

**Why B.** Series A returns to its old trend line after a shock, so the shock is temporary: A is trend-stationary, and you remove the fitted trend. Series B keeps the new level for good, so the shock is permanent: B has a unit root (difference-stationary), and you difference it.

$$\text{A: } X_t = a + bt + u_t \text{ (} u_t \text{ stationary)} \quad \text{B: } X_t = b + X_{t-1} + \varepsilon_t$$

**Why not the others.** A: differencing A over-differences it. C: the labels are swapped. D: removing a line from B leaves a random walk around the line, which is still not stationary.

**Takeaway.** Ask "does the shock fade or stay?" Fades → detrend. Stays → difference.

## Q9 · Answer A

**Why A.** Differencing a series that is already stationary creates artificial negative correlation. For white noise, for example:

$$\Delta \varepsilon_t = \varepsilon_t - \varepsilon_{t-1}, \quad \text{Corr}(\Delta \varepsilon_t, \Delta \varepsilon_{t-1}) = -0.5$$

The shared  $\varepsilon_{t-1}$  enters with opposite signs in neighbouring days, so a large negative lag-1 correlation appears.

**Why not the others.** B: momentum would show positive, not negative, correlation. C: differencing does not create a random walk. D: the pattern is caused by the extra difference, not by bad data.

**Takeaway.** A big negative lag-1 correlation after differencing is the warning sign of over-differencing.

### Q10 · Answer B

**Why B.** Each price wanders on its own (a unit root), but a combination  $Y - \beta X$  is stationary — it keeps coming back to a fixed level. That is the definition of cointegration: the walker and the dog both wander, but the leash keeps them close.

**Why not the others.** A: correlated returns move together day to day but can still drift apart over years; cointegration is about levels staying tied. C: each price wanders with permanent shocks, so it is not trend-stationary. D: white noise does not wander.

**Takeaway.** Cointegration = individually wandering series with a stationary gap.

### Q11 · Answer D

**Why D.** Measure how unusual today's spread is with a z-score:

$$S_t = Y_t - 1.5X_t = 106 - 90 = 16, \quad z = \frac{S_t - \bar{S}}{\sigma_S} = \frac{16 - 10}{2} = +3$$

The spread is far above normal, so Y is expensive relative to X. Sell the spread: sell 1 unit of Y and buy 1.5 units of X. If the spread returns toward 10, the trade earns money whichever way the overall market moves.

**Why not the others.** A: this bets with the move instead of against it. B: buying both is a market bet, not a pairs trade. C:  $z = +3$  is far outside the normal range.

**Takeaway.** Pairs trading: when the gap is far from normal, sell the rich leg, buy the cheap leg in the ratio  $\beta$ , and close as the gap returns.

### Q12 · Answer D

**Why D.** The whole trade rests on the spread being stationary. A merger changes the business link between the two companies, and the fresh check shows the cointegration is gone. Without the leash there is no force pulling the spread back to 10, so a  $z$  of +5 is not a stronger signal — the old mean is no longer meaningful. Close the trade and accept the loss.

**Why not the others.** A: adding to a position whose basis has disappeared enlarges the loss. B: spreads come back only while the cointegration holds. C: that is a new directional bet, not a pairs trade.

**Takeaway.** Every pairs trade needs an exit rule for when the relationship breaks (a stop-loss on  $z$  or a regular re-check of cointegration).

## Section II · Memory: AR, MA and Switching Between Them

### Q13 · Answer A

**Why A.** In an AR(1) a shock is carried forward through the series itself. Each day a share  $\phi$  of yesterday's effect survives, so the effect after  $h$  days is  $\phi^h$ :

$$\text{effect at } h = \phi^h : \quad 1, 0.5, 0.25, 0.125, \dots \quad (\phi = 0.5)$$

The echo halves every day but never hits exactly zero.

**Why not the others.** B: an MA(1) gives 1, 0.5, 0, 0, ... — it stops after one day. C: white noise has no echo (1, 0, 0, ...). D: a random walk keeps the full effect (1, 1, 1, ...).

**Takeaway.** AR memory fades gradually (geometric decay).

#### Q14 · Answer C

**Why C.** An MA(1) remembers only yesterday's shock:

$$X_t = \varepsilon_t + \theta \varepsilon_{t-1} \Rightarrow \text{effect: } 1, \theta, 0, 0, \dots$$

Two days later the shock is not in the equation at all, so its effect is exactly zero.

**Why not the others.** A, D: AR effects fade gradually and never cut off. B: a random walk never forgets.

**Takeaway.** MA memory has a fixed length: an MA(q) forgets a shock completely after q days.

#### Q15 · Answer B

**Why B.** The ACF at lag 2 is the plain correlation between Grandma and the Child. It counts everything, including the message that travelled through Mom. The PACF at lag 2 first removes what Mom passed on and then asks whether any direct line from Grandma to the Child remains.

**Why not the others.** A: the definitions are swapped. C: they differ whenever there is an indirect path. D: the PACF is used for any stationary series.

**Takeaway.** ACF = total link (direct + indirect). PACF = direct link only.

#### Q16 · Answer D

**Why D.** An MA(q) remembers shocks for exactly q days, so its ACF is non-zero up to lag q and zero after that. Here the ACF cuts off after lag 2, and the PACF fades gradually — the MA(2) pattern.

**Why not the others.** A: an AR(2) would show the opposite — PACF cut off at lag 2, ACF fading. B: white noise has no spikes. C: an ARMA would show both functions fading, not a clean cut.

**Takeaway.** The ACF cuts off → MA; the cut-off lag gives q.

#### Q17 · Answer B

**Why B.** For an AR(1), the ACF fades as  $\phi^k$  (here 0.7, 0.49, 0.34, ..., so  $\phi \approx 0.7$ ). Once yesterday is known, older days add nothing directly, so the PACF has one spike at lag 1 and then cuts off.

**Why not the others.** A: an MA(1) shows the ACF cutting off at lag 1. C: white noise has no spikes. D: a random walk's ACF stays near 1 and is not stationary.

**Takeaway.** The PACF cuts off → AR; the cut-off lag gives p.

| Pattern | ACF                  | PACF                 |
|---------|----------------------|----------------------|
| AR(p)   | fades gradually      | cuts off after lag p |
| MA(q)   | cuts off after lag q | fades gradually      |

| Pattern   | ACF             | PACF            |
|-----------|-----------------|-----------------|
| ARMA(p,q) | fades gradually | fades gradually |

### Q18 · Answer A

**Why A.** When neither function cuts off, the series has both kinds of memory. A small mixed model such as ARMA(1,1) can capture this with just two parameters. Start small and check the residuals.

**Why not the others.** B: a pure MA would need many terms, and its ACF would cut off. C: a random walk is not stationary. D: this is exactly the case ARMA models were built for.

**Takeaway.** Both fade → try a small ARMA first.

### Q19 · Answer C

**Why C.** With  $\phi = 1.2$  each shock grows instead of fading: 1, 1.2, 1.44, 1.73, ... The series explodes, so it cannot be stationary.

$$\text{AR}(1) \text{ is stationary} \iff |\phi| < 1$$

**Why not the others.** A: a larger  $\phi$  means a weaker pull, and beyond 1 the pull becomes a push. B, D: the model type does not change.

**Takeaway.**  $|\phi| < 1$  fades,  $\phi = 1$  never fades (unit root),  $|\phi| > 1$  explodes.

### Q20 · Answer C

**Why C.** Replace yesterday's value with its own equation, again and again:

$$\begin{aligned} X_t &= \varepsilon_t + 0.8X_{t-1} \\ &= \varepsilon_t + 0.8(\varepsilon_{t-1} + 0.8X_{t-2}) = \varepsilon_t + 0.8\varepsilon_{t-1} + 0.64X_{t-2} \\ &= \varepsilon_t + 0.8\varepsilon_{t-1} + 0.64\varepsilon_{t-2} + 0.512\varepsilon_{t-3} + \dots \end{aligned}$$

The leftover term  $0.8^h X_{t-h}$  shrinks to zero because  $|0.8| < 1$ . What remains is an MA with endless terms and weights  $\phi^j = 1, 0.8, 0.64, 0.512, \dots$

**Why not the others.** A: one term is not enough; the substitution never stops. B: it can be rewritten — that is the point of the question. D: the weights shrink as powers of 0.8; they are not all equal.

**Takeaway.** A stationary AR(1) is an MA( $\infty$ ) with weights  $\phi^j$  (the Wold form).

### Q21 · Answer D

**Why D.** In the MA( $\infty$ ) form, the weight on  $\varepsilon_{t-j}$  is exactly how much of the shock from  $j$  days ago is still inside today's value (the impulse response). With  $\phi = 0.8$ , after 3 days 0.512 of a shock remains, after 10 days about 0.11. The weights must shrink toward zero for the series to stay stationary; if they did not ( $\phi = 1$ ), old shocks would pile up as in Q5.

**Why not the others.** A: the MA( $\infty$ ) has endless weights, not fewer parameters — it is still governed by the one  $\phi$ . B: an AR is easy to estimate. C: rewriting a model does not change the data; a non-stationary

AR does not even have this form.

**Takeaway.** AR → MA answers the question "how long does a shock last?" and gives a stationarity check.

## Q22 · Answer B

**Why B.** Solve the MA(1) for the shock and substitute repeatedly:

$$\begin{aligned}\varepsilon_{t-1} &= X_{t-1} - 0.5\varepsilon_{t-2} = X_{t-1} - 0.5X_{t-2} + 0.25X_{t-3} - \dots \\ X_t &= \varepsilon_t + 0.5\varepsilon_{t-1} = 0.5X_{t-1} - 0.25X_{t-2} + 0.125X_{t-3} - \dots + \varepsilon_t\end{aligned}$$

The weights are  $-(\theta)^j$ : 0.5, -0.25, 0.125, ... They alternate in sign because each substitution brings in another factor of  $-\theta$ , and they shrink because  $|\theta| < 1$ .

**Why not the others.** A: a single AR term drops all the older corrections. C: past shocks can be written in terms of past  $X$ , so it can be done. D: growing weights would happen only with  $|\theta| > 1$ , which is the non-invertible case.

**Takeaway.** An invertible MA(1) is an AR( $\infty$ ) with alternating, shrinking weights.

## Q23 · Answer A

**Why A.** Shocks are never observed — only  $X$  is. To forecast tomorrow from an MA(1) we need today's shock  $\varepsilon_t$ , and the AR( $\infty$ ) form recovers it from the data we have:

$$\varepsilon_t = X_t - 0.5X_{t-1} + 0.25X_{t-2} - 0.125X_{t-3} + \dots$$

This works only when  $|\theta| < 1$  (invertibility). Otherwise the weights on old data grow, and the distant past would dominate the estimate of today's shock.

**Why not the others.** B: a finite MA is always stationary already. C: an MA has memory; it is just short. D: the rewrite adds no new parameters; everything still comes from  $\theta$ .

**Takeaway.** MA → AR turns hidden shocks into something computable from data, which makes forecasting possible. Invertibility ( $|\theta| < 1$ ) is the condition.

## Q24 · Answer C

**Why C.** Q20–Q23 show that the same process can be written either way. A stationary AR(1) is an MA( $\infty$ ), and an invertible MA(1) is an AR( $\infty$ ). The data and the memory are the same; only the description changes. So choose the form that needs the fewest parameters (parsimony): a series whose ACF cuts off is described more cheaply as an MA, and one whose PACF cuts off as an AR.

**Why not the others.** A, B: neither form is better in general, and MA models are for stationary data. D: rewriting changes the description, not the data.

**Takeaway.** AR and MA are two languages for one memory. Pick the shorter sentence.

## Q25 · Answer B

**Why B.** For a stationary AR(1), today's distance from the mean shrinks by a factor  $\phi$  for each step ahead:

$$\hat{X}_{t+h} = \mu + \phi^h(X_t - \mu) \longrightarrow \mu \quad (h \rightarrow \infty)$$

The forecast moves from 4% back toward 2%. The uncertainty band widens with the horizon but levels off at the series' own variance.

**Why not the others.** A: that is the random-walk forecast. C: the return is gradual, not a jump (unless  $\phi = 0$ ). D: nothing pushes a stationary series further away.

**Takeaway.** Stationary forecasts revert to the mean, like a stretched rubber band.

#### Q26 · Answer A

**Why A.** If a model has captured all the memory, its residuals should look like white noise, with no ACF spikes. A spike at lag 1 means some one-day memory is still unexplained. Revise the model (for example, add an AR or MA term) and look at the residual ACF again.

**Why not the others.** B, D: leftover structure means the model is incomplete. C: the spike is about the residuals of this model, not proof that returns can be predicted in general.

**Takeaway.** Always check the residuals: what is left over should be white noise.

### Section III · Trends and Seasons: ARIMA and SARIMA

#### Q27 · Answer D

**Why D.** ARMA needs a stationary series, and a staircase-like trend is not stationary. The "I" (integrated) step works in three moves: difference the series  $d$  times until it is stationary, fit an ARMA to those changes, then add the forecast changes back up to get a forecast of the level.

$$\text{ARIMA}(p, 1, q) : \quad \Delta X_t = X_t - X_{t-1} \sim \text{ARMA}(p, q), \quad \hat{X}_{t+h} = X_t + \sum_{i=1}^h \widehat{\Delta X}_{t+i}$$

**Why not the others.** A: the seasonal part is the S in SARIMA. B: outlier cleaning is a separate step. C: a mean-reverting forecast is what a stationary ARMA does, not the "I".

**Takeaway.** I = difference  $\rightarrow$  model the changes  $\rightarrow$  add them back up.

#### Q28 · Answer B

**Why B.** If the first difference is white noise, then  $\Delta X_t = \varepsilon_t$ : one difference ( $d = 1$ ), no AR terms ( $p = 0$ ), no MA terms ( $q = 0$ ). That is ARIMA(0,1,0), the random walk.

**Why not the others.** A: ARIMA(1,0,0) is a stationary AR(1) of the level. C: ARIMA(0,0,1) is an MA(1) of the level. D: a second difference is not needed and would over-difference.

**Takeaway.** Random walk = ARIMA(0,1,0). Read the order as (AR terms, differences, MA terms).

#### Q29 · Answer A

**Why A.** Model P is stationary, so its forecast reverts to the long-run mean (as in Q25). Model Q has a unit root with drift  $\mu$ , so its forecast starts from today's price and moves along the drift. Its uncertainty grows

with the horizon without limit.

$$P: \hat{X}_{t+h} \rightarrow \mu \quad Q: \hat{X}_{t+h} = X_t + h\mu, \quad \text{Var} = h\sigma^2$$

**Why not the others.** B: a unit-root series has no average to return to. C: P reverts, and Q moves with the drift. D: the behaviours are swapped.

**Takeaway.** Stationary forecasts go home; unit-root forecasts carry on from where they are, with a widening band.

### Q30 · Answer C

**Why C.** One difference already gives a stationary series. The second difference produces the signs of over-differencing from Q9: a large negative lag-1 correlation and a bigger variance. Use the smallest d that achieves stationarity.

**Why not the others.** A: the original series is not stationary. B: the second difference is one too many. D: even further from the right answer.

**Takeaway.** Difference only as much as needed. Too little leaves a unit root; too much adds false negative correlation and noise.

### Q31 · Answer D

**Why D.** Monthly data with spikes at lags 12, 24 and 36 means each month resembles the same month in earlier years. That is a yearly season with period  $s = 12$ .

**Why not the others.** A: a random walk shows a slow, smooth decay, not spikes at multiples of 12. B: over-differencing shows a negative spike at lag 1. C: white noise has no spikes.

**Takeaway.** ACF spikes at  $s, 2s, 3s \rightarrow$  seasonality with period  $s$  (12 for monthly data, 4 for quarterly).

### Q32 · Answer A

**Why A.** Comparing each month with the same month a year earlier is a seasonal difference:

$$(1 - L^{12})X_t = X_t - X_{t-12}$$

The summer peak appears in both this June and last June, so it cancels. What remains is the change that is not explained by the season.

**Why not the others.** B: a regular first difference ( $X_t - X_{t-1}$ ) compares neighbouring months and leaves the yearly pattern in place. C: detrending removes a straight line, not a repeating pattern. D: a moving-average term models shocks; it does not remove seasonality.

**Takeaway.** A regular difference removes a trend; a seasonal difference removes a repeating season. SARIMA can use both.

## Section IV · Forecasting the Storm: ARCH, GARCH and VaR

### Q33 · Answer C

**Why C.** The returns' ACF is near zero, so yesterday's direction tells you nothing about today's direction. The squared returns' ACF is clearly positive, so a big move (up or down) yesterday makes a big move today more likely. This is volatility clustering: calm and stormy periods come in runs.

**Why not the others.** A: the direction is not predictable. B: returns show no unit-root sign. D: the two ACFs measure different things — direction and size — so they need not agree.

**Takeaway.** Returns: no memory in direction, strong memory in size. This is the reason for ARCH and GARCH.

#### Q34 · Answer D

**Why D.** The unconditional variance is a single long-run number for the whole sample. The conditional variance is today's variance given what happened recently, so it changes every day:

$$\text{unconditional: } \text{Var}(r_t) = \bar{\sigma}^2 \quad \text{conditional: } \sigma_t^2 = \text{Var}(r_t \mid r_{t-1}, r_{t-2}, \dots)$$

A series can have a constant unconditional variance (so it is stationary) while its conditional variance moves up in storms and down in calm periods.

**Why not the others.** A: they differ whenever volatility clusters. B: the definitions are swapped. C: both apply to returns.

**Takeaway.** GARCH models the conditional variance; its long-run level is the unconditional variance.

#### Q35 · Answer B

**Why B.** The shock enters squared, so its sign disappears:

$$(-0.05)^2 = (+0.05)^2 = 0.0025 \quad \Rightarrow \quad \sigma_t^2 = \omega + \alpha \times 0.0025 \text{ in both cases}$$

A 5% fall and a 5% rise raise today's variance by exactly the same amount.

**Why not the others.** A: a rise does not lower the variance; any large move raises it. C: both have an effect. D: the sign does not matter in a basic ARCH.

**Takeaway.** In ARCH and basic GARCH only the size of yesterday's move matters. (Models that let falls matter more are an extension beyond this course.)

#### Q36 · Answer A

**Why A.** In GARCH(1,1),  $\omega$  is a constant floor,  $\alpha$  is the reaction to yesterday's surprise, and  $\beta$  multiplies yesterday's variance. So  $\beta$  carries yesterday's risk level into today. A large  $\beta$  means a stormy (or calm) period tends to continue.

**Why not the others.** B: the shock is squared, so the sign does not enter. C: the average return belongs to the mean equation, not the variance equation. D: detrending prices has nothing to do with GARCH.

**Takeaway.**  $\alpha$  = reaction to news;  $\beta$  = memory of past risk.

#### Q37 · Answer D

**Why D.** The sum  $\alpha + \beta$  (persistence) is the share of extra volatility that survives each day. The half-life is the number of days until half of a spike has faded:

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)} : \quad 0.95 \Rightarrow \approx 13.5 \text{ days}, \quad 0.99 \Rightarrow \approx 69 \text{ days}$$

In trading days, that is about two to three weeks for A and over three months for B. B's storm lingers much longer.

**Why not the others.** A: different persistence means different speeds. B: it is the reverse — the closer to 1, the slower the fade. C: both fade, because both sums are below 1.

**Takeaway.**  $\alpha + \beta$  near 1 = long-lasting volatility. The persistence plays the same role as  $\varphi$  in an AR(1).

### Q38 · Answer B

**Why B.** When  $\alpha + \beta < 1$ , volatility is mean-reverting. The forecast moves step by step from today's high level toward the long-run level:

$$\hat{\sigma}_{t+h}^2 = \bar{\sigma}^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \bar{\sigma}^2), \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$$

**Why not the others.** A: that would need  $\alpha + \beta = 1$ . C: the fall is gradual, and variance never reaches zero ( $\omega > 0$ ). D: growth without limit would need  $\alpha + \beta > 1$ .

**Takeaway.** Volatility forecasts revert to their long-run level — the same rubber-band idea as Q25, applied to risk.

### Q39 · Answer C

**Why C.** A 99% one-day VaR of 2% is a threshold: the daily loss should exceed 2% on only about 1% of days.

$$P(\text{loss} > \text{VaR}_{99\%}) = 1\%$$

Over 250 trading days, about 2 to 3 breaks are expected. VaR says nothing about how large a loss is once the threshold is crossed.

**Why not the others.** A: VaR is not a maximum; larger losses happen on about 1% of days. B: "exactly 2% on one day in a hundred" misreads the threshold as a fixed loss. D: VaR is about the tail, not the average.

**Takeaway.** VaR = how bad a normal bad day gets, not the worst case.

### Q40 · Answer C

**Why C.** A VaR built on one fixed volatility uses the same threshold in calm and stormy times. When volatility jumps in a crisis, the fixed threshold is far too tight, so breaks pile up in that one month. A GARCH VaR uses today's conditional volatility, so it widens when the storm arrives:

$$\text{VaR}_t = z_{0.99} \sigma_t \approx 2.33 \sigma_t \quad (\text{fixed VaR uses one } \bar{\sigma} \text{ instead of } \sigma_t)$$

A good VaR should have about 1% breaks, scattered over time rather than bunched together.

**Why not the others.** A: portfolio size does not cause clustered breaks. B: the mean return is not the issue; the volatility is. D: bunched breaks are the warning sign that the model ignores changing risk.

**Takeaway.** Risk changes over time, so the VaR must change with it. That is the practical payoff of GARCH.